

Absolute figure and table placement with the **absfigure** package

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NOTE This is version 0.1 of `absfigure`. The package is functional but should be considered experimental. Please report any issues.

1 Introduction

L^AT_EX’s float mechanism (`\begin{figure}[t/b/h/p]`) provides only *relative* placement hints. There is no built-in way to say “put this figure at the bottom of page 6, column 2.” The `absfigure` package provides exactly that capability for both figures and tables through a simple key-value interface.

Floats placed with `absfigure` integrate with L^AT_EX’s page builder: text correctly reflows around them, `\caption`, `\label`, and `\ref` all work normally, and ordinary floats continue to function alongside absolute ones. Both `\listoffigures` and `\listoftables` pick up the placed floats automatically.

1.1 Requirements

- L^AT_EX 2_ε (1995/12/01 or later)
- The `keyval` package (part of the `graphics` bundle)
- The `environ` package (available on CTAN)

1.2 Loading the package

```
\usepackage{absfigure}
```

No package options are currently defined.

2 User interface

2.1 The `absfigure` environment

```
\begin{absfigure}{\langle keys \rangle}
\langle figure body \rangle
\end{absfigure}
```

Places a single-column figure on a specific page and column. The *⟨figure body⟩* may contain `\includegraphics`, `\caption`, `\label`, and any other content you would normally put inside a L^AT_EX figure environment.

EXAMPLE

```
\begin{absfigure}{page=6, pos=b, col=2}
\includegraphics[width=\columnwidth]{fig.pdf}
\caption{My figure}
\label{fig:myfig}
\end{absfigure}
```

2.2 The `absfigure*` environment

```
\begin{absfigure*}{\langle keys \rangle}
\langle figure body \rangle
\end{absfigure*}
```

Places a spanning (double-column) figure, analogous to L^AT_EX’s `figure*`. The `col` key is ignored for this environment since the figure spans both columns.

EXAMPLE

```
\begin{absfigure*}{page=3, pos=t}
  \includegraphics[width=\textwidth]{wide.pdf}
  \caption{A wide figure}
\end{absfigure*}
```

2.3 The abstable environment

```
\begin{abstable}{\langle keys \rangle}
  \langle table body \rangle
\end{abstable}
```

Places a single-column table on a specific page and column. The *table body* may contain `tabular`, `\caption`, `\label`, and any other content you would normally put inside a L^AT_EX table environment. The same keys (`page`, `pos`, `col`) are accepted.

EXAMPLE

```
\begin{abstable}{page=4, pos=t, col=1}
  \centering
  \caption{My table}
  \label{tab:mytab}
  \begin{tabular}{lcc}
    \hline
    Item & Value & Unit \\
    \hline
    Alpha & 1.0 & m/s \\
    \hline
  \end{tabular}
\end{abstable}
```

2.4 The abstable* environment

```
\begin{abstable*}{\langle keys \rangle}
  \langle table body \rangle
\end{abstable*}
```

Places a spanning (double-column) table, analogous to L^AT_EX's `table*`. The `col` key is ignored for this environment since the table spans both columns.

EXAMPLE

```
\begin{abstable*}{page=5, pos=b}
  \centering
  \caption{A wide table}
  \begin{tabular}{lccccc}
    \hline
    Exp & T1 & T2 & T3 & Mean & Std \\
    \hline
    A & 4.2 & 4.5 & 4.1 & 4.27 & 0.21 \\
    \hline
  \end{tabular}
\end{abstable*}
```

3 Keys

3.1 `page`

`page=<integer>` — Target page number (positive integer).
`page=current` — Place on the page currently being built (default).
`page=end` — Place on an extra page appended after the document.

If `page` is omitted, it defaults to `current` — the float is placed on the page where the environment appears in the source.

An error is raised if `page` is zero, negative, or non-numeric (other than the special values `current` and `end`).

If the target page number exceeds the number of pages in the document, the float is never placed and a hard error is raised at the end of the $\text{\LaTeX}$ run.

WARNING When using an explicit page number, the float must be defined in the source *before* its target page is shipped out. In practice, group such floats right after `\begin{document}`. With `page=current` (or the default), the float is placed on the page being built where the environment appears, so ordering is not a concern.

3.2 `pos`

`pos=t` — Place at the top of the page (default).
`pos=b` — Place at the bottom of the page.

The aliases `pos=top` and `pos=bottom` are also accepted.

An error is raised if `pos` is set to anything other than `t/top` or `b/bottom`.

NOTE The $\text{\LaTeX}$ kernel does not natively support bottom-placed spanning floats (there is no `\@dblbotlist`). The `absfigure` package implements its own spanning-bottom mechanism, so `pos=b` works with all four environments (`absfigure`, `absfigure*`, `abstable`, `abstable*`).

3.3 `col`

`col=1` — Place in the first (left) column (default).
`col=2` — Place in the second (right) column.

The aliases `col=l/col=left` and `col=r/col=right` are also accepted.

This key is only meaningful in `\twocolumn` mode. In single-column documents, `col=2` (or `col=right`) raises an error.

For starred environments (`absfigure*`, `abstable*`), this key is silently ignored (the float always spans both columns).

An error is raised if `col` is set to anything other than `1/left` or `2/right`.

4 Captions, labels, and cross-references

`\caption` and `\label` work normally inside all four environments. They are executed at *placement* time (during the output routine), so:

- The figure/table counter is incremented in the order floats are actually placed, not the order they appear in the source.
- `\ref` and `\pageref` resolve correctly after two compilations (standard $\text{\LaTeX}$ behaviour for cross-references).
- Figures appear in `\listoffigures` and tables in `\listoftables` automatically.

5 Error handling

The package performs validation on all keys and issues errors or warnings as appropriate:

page omitted Defaults to `current` (the page being built). No error.

page is zero, negative, or non-numeric **Error:** Invalid page value ‘ $\langle value \rangle$ ’. The `page` key requires a positive integer, `current`, or `end`.

page exceeds total pages **Error** at end of run: Float N (figure/table, target page=X) was never placed. Compilation is interrupted; no float is silently dropped.

page=current Valid. Resolves to the page number being built at definition time.

page=end Valid. An extra page is appended after the document’s last page, and the float is placed there.

pos is not t/top or b/bottom **Error:** Invalid pos value ‘ $\langle value \rangle$ ’.

col is not 1/l/left or 2/r/right **Error:** Invalid col value ‘ $\langle value \rangle$ ’.

col=2 or col=right in single-column mode **Error:** col=2 (right) used in single-column mode.

6 How it works

Understanding the internal mechanism may be helpful for advanced users and for debugging.

6.1 Core mechanism

L^AT_EX calls `\@startcolumn` at the start of every column (and page). The package patches this command to inject absolute figures *before* deferred floats are processed:

```
\@startcolumn (patched):
1. \global\@colroom\@colht      % reset height
2. detect new column vs re-entry
3. \absf@inject@figures          % OUR HOOK
4. \@colroom -= \absf@colreduce  % apply reduction
5. \@tryfcolumn / deferred      % original logic
```

6.2 Re-entry handling

The kernel may call `\@startcolumn` multiple times for the same column (via `\@whilesw@if@fcolmade`). Each re-entry resets `\@colroom`, undoing any reductions. The package tracks the current column identity (page number + column flag) and stores the cumulative space reduction in a dimension register (`\absf@colreduce`). On re-entry, injection is skipped but the stored reduction is re-applied.

6.3 Injection logic

For each pending float, the package checks whether the current page counter (`\c@page`) matches the target page and whether `\if@firstcolumn` matches the target column. On a match:

1. A float box register is allocated from `\@freelist`.
2. The captured body tokens are typeset inside it (with `\@captype` set to `figure` or `table` so `\caption/\label` work and counters increment correctly).

3. `\count\@currbox` is set to encode the float type and placement using the kernel's `\ftype@⟨type⟩` multiplier.
4. The box is added to `\@toplist` or `\@botlist` and the space reduction is accumulated in `\absf@colreduce`.

For spanning floats (`absfigure*`, `abstable*`), the box is added to `\@dbltoplist` and `\@colht` is reduced instead (this reduction naturally survives re-entries since `\@startcolumn` resets `\@colroom` from `\@colht`).

6.4 Why text reflows correctly

Reducing `\@colroom` inside `\@startcolumn` propagates to `\vsize`, which the $\text{\TeX}$ page builder uses to determine where to break the column. Less available height means less text on the page, leaving room for the figure.

7 Limitations and known issues

- Floats must be defined in the source *before* their target page. The recommended practice is to group all `absfigure`/`abstable` environments right after `\begin{document}`.
- If many absolute floats target the same page, the available text height may become negative and $\text{\TeX}$ will produce an overfull vbox. Plan float sizes accordingly.
- Float numbering follows placement order, not source order. If you define figure A (targeting page 5) before figure B (targeting page 2), figure B will get a lower number. The same applies to tables.
- Cross-references (`\ref`, `\pageref`) require two compilations to resolve, as is standard for $\text{\LaTeX}$.
- The package has not been tested with the `multicol` package. It is designed for $\text{\LaTeX}$'s built-in `\twocolumn` mechanism.

8 Complete example

```
\documentclass[twocolumn]{article}
\usepackage{graphicx}
\usepackage{lipsum}
\usepackage{absfigure}

\begin{document}

% Define figures up front
\begin{absfigure}{page=2, pos=t, col=1}
  \centering
  \includegraphics[width=\columnwidth]{fig1.pdf}
  \caption{Top of page 2, column 1}
  \label{fig:one}
\end{absfigure}

\begin{absfigure*}{page=3, pos=t}
  \centering
  \includegraphics[width=\textwidth]{wide.pdf}
  \caption{Spanning figure on page 3}
  \label{fig:wide}
\end{absfigure*}
```

```

\end{absfigure*}

% A table on page 2, bottom of column 2
\begin{abstable}{page=2, pos=b, col=2}
  \centering
  \caption{Results}
  \label{tab:results}
  \begin{tabular}{lcc}
    \hline
    Method & Accuracy & F1 \\
    \hline
    Ours & 94.2 & 0.91 \\
    \hline
  \end{tabular}
\end{abstable}

\begin{absfigure}{page=end}
  \centering
  \includegraphics[width=\columnwidth]{extra.pdf}
  \caption{Appended at the end}
  \label{fig:extra}
\end{absfigure}

See Figure~\ref{fig:one}, Table~\ref{tab:results},
and Figure~\ref{fig:extra}.

\lipsum[1-30]

\end{document}

```

9 Version history

v0.1 (2026/02/25) Initial release.

- `absfigure`, `absfigure*`, `abstable`, and `abstable*` environments
- Keys: `page`, `pos`, `col`
- `page=current` (default when `page` omitted)
- `page=end` support
- Validation of all keys with meaningful error messages
- End-of-document warnings for unplaced figures